

Roadmap to a Full App

What exists today is a working **v1 proof-of-concept**: the whole loop runs end to end (search → cited answer → comment-to-DM → Demand Radar → affiliate links), with 42 passing tests against real Postgres+pgvector. This document lists **everything that still needs to be added** to turn that POC into a complete, multi-tenant, self-serve product.

Each item is tagged by priority: - **P0** — required before any real external launch - **P1** — needed to scale past the first hand-held creator - **P2** — growth, polish, and the monetization expansion ladder

Legend for status: **stub** = placeholder exists, **missing** = not started, **partial** = works but not production-grade.

1. Onboarding & Authentication

Item	Priority	Status	Notes
Instagram OAuth sign-in flow (UI)	P0	missing	Today tokens are pasted via CLI/admin. Need the real "Connect Instagram" button → consent → callback → token exchange. The token-exchange helpers already exist in <code>instagram.py</code> .
YouTube OAuth flow	P1	missing	Same pattern for owned/authorized channels.
Long-lived token refresh job	P0	missing	IG long-lived tokens expire (~60 days); need a scheduled refresh before expiry or creators silently disconnect.
Token encryption at rest	P0	partial	Tokens are stored in plaintext columns. Encrypt with a KMS/secret key.
Creator self-serve signup + onboarding wizard	P1	missing	Guided flow: connect → backfill progress → label questions → publish link.
Creator login / session management	P0	missing	Real creator accounts (email+password or social login), not a single shared admin bearer token.
Role-based access (creator vs operator vs admin)	P1	missing	Current admin API is one global token.

2. Multi-Tenancy, Billing & Plans

Item	Priority	Status	Notes
Subscription billing (Stripe)	P0	missing	Creator (~\$49-99/mo) and Pro (~\$149-299/mo) tiers from the plan. Checkout, webhooks, dunning.
Plan-based feature gating	P1	missing	e.g. comment-to-DM + storefront search are Pro-only.
Usage metering & quotas	P1	missing	Cap AI spend per creator; meter searches/DMs for fair-use and cost control.
Free-trial / paid-pilot logic	P1	missing	Convert pilots to paid once Radar shows value.
Per-creator custom domains	P2	missing	Plan defers this; path-based <code>/<handle></code> ships first.

3. Creator Dashboard (web UI)

The admin surface is API-only today. Creators need a real interface.

Item	Priority	Status	Notes
Demand Radar review UI	P0	missing	Evidence cards with "useful / not / made" buttons. API exists; no screen.
DM approval inbox	P0	missing	Approve/edit/reject queued replies. API exists; no screen.
Content/corpus browser	P1	missing	See ingested posts, transcript/OCR coverage, re-trigger processing, delete.
Product & affiliate manager	P1	missing	Add/approve products, set affiliate links, map to posts.
Metrics dashboard	P1	partial	Metrics endpoint exists; needs charts (searches, click-rate, no-answer rate, ROI).
Distribution setup checklist	P2	missing	Guide the bio link + "Search" Highlight + pinned reel behaviour change.

4. Retrieval & Answer Quality (the moat)

The POC uses offline "fake" AI providers by default. Production quality depends on:

Item	Priority	Status	Notes
Wire real LLM + embedding providers	P0	partial	Interfaces + Anthropic/OpenAI adapters exist; need real keys, model selection, prod config.
Real transcription (Whisper) at scale	P0	partial	Adapter exists; needs GPU/worker capacity and a media-download retry path (IG CDN URLs expire).
Real OCR (PaddleOCR) at scale	P1	partial	Adapter exists; tune keyframe sampling and cost.
Reranking model	P1	missing	Plan calls for a rerank stage; today it's RRF fusion only. Add a cross-encoder/LLM reranker.
Confidence-threshold calibration	P0	partial	Thresholds are hand-set; calibrate against the labelled set per creator/niche.
Larger evaluation sets + CI gate	P0	partial	Harness exists; need 40–60 labelled Qs per creator wired into CI to block regressions.
Freshness & contradiction handling	P1	missing	When newer posts supersede old advice, prefer/flag the current one.
Multilingual support	P2	missing	Swap to a multilingual embedding model + language-aware FTS config.
Creator-voice answer styling	P2	missing	Plan keeps this secondary; layer voice on top of the correct, cited answer.

5. Comment-to-DM Automation

Item	Priority	Status	Notes
Production webhook subscription setup	P0	partial	Receiver + signature check exist; need the Meta webhook subscription wired per creator.
Graduated auto-send (post-gate)	P1	partial	Approval mode works; add the automatic path once the citation gate passes, per-post opt-in.
Per-post trigger CTA management	P1	missing	UI to set "Comment PLAN..." triggers per post.
Rate-limit queue durability	P1	partial	Hourly cap enforced in-process; make the overflow queue durable across restarts.
DM delivery analytics	P2	partial	Events logged; surface send-success and click-through.

6. Demand Radar & Insights

Item	Priority	Status	Notes
Weekly digest email delivery	P0	stub	Currently prints to a log. Wire a real email provider (SES/Postmark/Resend).
Digest via DM option	P2	missing	Plan offers email or DM.
Better clustering quality	P1	partial	Greedy centroid clustering is fine for v1; evaluate HDBSCAN/ embeddings tuning at volume.
Insight → draft generator (v2)	P2	missing	One-tap reel script/hook/caption from a Radar card. Premium tier.
Sponsor pitch reports (v2)	P2	missing	Auto one-pager: "412 questions about X this month."

7. Monetization Expansion Ladder (plan \$13)

Item	Priority	Status	Notes
Affiliate mapping + click tracking	DONE	—	Already built (v1 proof of ROI).
Email capture bridge (v1.5)	P2	missing	"Get the full checklist by email" on answer cards + DMs; builds the creator's owned list.
Storefront search (v1.5)	P2	missing	Mark premium content; "covered in depth in module 3 →" teaser → checkout.
Embeddable search widget (v1.5)	P2	missing	The search box on the creator's own site + "Powered by" footer (referral loop).
Paid Q&A on "no answer" (v2)	P2	missing	Turn the failure state into a paid question / coaching funnel.

8. Infrastructure & Operations

Item	Priority	Status	Notes
Production deployment (cloud)	P0	partial	docker-compose exists for local; need managed Postgres, object storage, container hosting, TLS, a domain.
Database migrations tool (Alembic)	P0	partial	Today schema is create-from-metadata. Adopt versioned migrations before live data exists.
Secrets management	P0	missing	Move keys/tokens out of .env into a secrets manager.
CI/CD pipeline	P0	missing	Run tests + eval gate + lint on every push; automated deploys.
Background-job reliability	P1	partial	RQ + scheduler exist; add retries, dead-letter handling, and a durable scheduler (not the in-process loop).
Monitoring, logging, alerting	P0	missing	Error tracking (Sentry), uptime checks, AI-cost dashboards, queue depth alerts.
Event analytics pipeline	P1	partial	Events table exists; wire PostHog or a warehouse for product analytics.
Backups & disaster recovery	P0	missing	Automated DB backups + restore drills; media storage durability.
Rate limiting & abuse protection	P1	missing	Throttle the public search API; bot protection.
Caching / CDN for the fan page	P1	missing	Cache result pages; serve the Qwik app via CDN for in-app-browser speed.

9. Legal, Privacy & Compliance

Item	Priority	Status	Notes
Privacy policy + ToS	P0	missing	Required by Meta App Review and GDPR.
GDPR/CCPA deletion & export	P0	partial	Audience data is pseudonymized and post-deletion purges chunks; need a user-facing deletion/export request path.
Meta App Review submission	P0	missing	Advanced Access for external creators; request only scopes the demo uses.
Data processing agreements	P1	missing	DPA's with AI/infra subprocessors.
Content moderation / safety guardrails	P1	partial	"No answer" + citation gate reduce hallucination; add abuse/PII filters on audience input.
Cookie consent (where required)	P2	missing	If analytics cookies are used in applicable regions.

10. Quality, Testing & Docs

Item	Priority	Status	Notes
Backend test suite	DONE	—	42 tests against real Postgres+pgvector.
Frontend tests (web app)	P1	missing	Component + end-to-end tests for the Qwik pages.
Load / performance testing	P1	missing	Validate search latency and webhook throughput on a viral post.
API documentation	P2	partial	FastAPI auto-docs exist; add a published API reference if exposing it.
Runbooks / on-call docs	P1	missing	How to handle token expiry, Meta outages, cost spikes.

Suggested Sequencing

Phase A — make it real for one creator (mostly P0) Real AI keys → cloud deployment + managed Postgres → migrations → secrets → monitoring/backups → privacy policy. (This matches `GOING_LIVE.md`.)

Phase B — let creators run themselves (P0/P1) OAuth sign-in + token refresh → creator login & dashboard (Radar review + DM inbox) → email digest delivery → Stripe billing → Meta App Review.

Phase C — scale & differentiate (P1/P2) Reranking + freshness → plan gating & quotas → CI/CD + analytics → the monetization ladder (email capture, storefront, widget) → draft generator & sponsor reports.

Companion documents: `IMPLEMENTATION_PLAN.md` (the technical v1 plan), `GOING_LIVE.md` (plain-language launch checklist), `README.md` (how to run it).